

MODULE 25

# SQL vs NoSQL and Future of Databases

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DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

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## SQL vs NoSQL

SQL: fixed schema, ACID, complex queries  
MySQL, PostgreSQL, Oracle, SQL Server

NoSQL: flexible schema, horizontal scale  
MongoDB, Redis, Cassandra, DynamoDB

SQL: banking, ERP, e-commerce  
NoSQL: social media, IoT, real-time analytics

## NewSQL Databases

Combines SQL ACID guarantees + NoSQL scalability:

Google Spanner: globally distributed SQL  
CockroachDB: resilient distributed SQL  
TiDB: MySQL-compatible distributed  
YugabyteDB: high-performance distributed

## Cloud Databases

AWS RDS: managed MySQL, PostgreSQL, Oracle  
AWS Aurora: 5x faster MySQL compatible  
Google BigQuery: serverless analytics warehouse  
Azure SQL: managed SQL Server  
PlanetScale: serverless MySQL

## OLTP vs OLAP

OLTP: many small transactions, normalized, fast reads/writes  
Examples: MySQL, PostgreSQL

OLAP: complex analytics, denormalized, large scans  
Examples: BigQuery, Snowflake, Redshift

ETL: Extract from OLTP, Transform, Load into OLAP

## Career Paths

SQL opens paths to:

Database Administrator (DBA)

Data Analyst

Data Engineer

Data Scientist

Backend Developer

Next steps: Python, data modeling,  
certifications, Kaggle datasets,  
data warehouses (Snowflake, BigQuery)